Imperial College London

AERO96005 - GROUP DESIGN PROJECT

Wholly Reusable Launch System (Recovery)

Aerodynamic Design and Analysis of Empennage and Rotor Tip of Synchropter Concept for the Aerial-Recovery Mission

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Abstract

This report present the conceptual and preliminary design phase of empennage and rotor blade for the proposed synchropter concept. Pros and Cons of different rotor and empennage configurations are discussed, where a synchropter with H-tail is selected as the baseline configuration. Based on 2-D CFD analysis, NACA63206 is selected as the aerofoil for rotor blade tip due to excellent performance in terms of M_{DD} and L/D_{max}. The lift and drag polars for both root and tip aerofoils are obtain through CFD. Inspired by the BERP-rotor, the rotor tip geometry employs sweep and notch to reduce wave drag and to delay stall. Empennage surfaces are sized according to empirical relations based on past helicopter data. The performance analysis of empennage includes a preliminary downwash model for the rotor-tail interaction. A CFD analysis of the grapple device is conducted to provide relevant information for the System team. Finally, directions of future developments and potential improvements are discussed.

Acknowledgements

To Errikos - for providing this amazing project to his disciples
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Part I
Conceptual Design

During the first and second week of the project, the conceptual design phase of the helicopter concept was conducted in the Aerodynamics sub-team focusing on the rotor, fuselage and empennage design. This section details the trade-off analysis and decision making in the first two weeks.

1 Rotor Configuration

1.1 Mission definition

The definition of mission profile depends largely on the Launcher team. In the first and second week of the project, the size of the payload and location of reentry was known only as a range of 1 to 4 tons and within 50km radius on the sea. The altitude of recovery maneuver was set at 10,000 ft at a relatively low speed. These reference values are the starting point of the helicopter conceptual design.

1.2 Design drivers

The key design drivers for the helicopter rotor configuration is summarised in Table 1.

Design driver	Abbreviation	Implications
Maneuverability	M.	To ensure safe recovery especially at low speed
Stability	S.	Easier controller design and reduced pilot workload
Cruise performance	CP.	Ability to travel to destination quickly
Hovering performance	HP.	Efficient standstill &. high payload capacity
Life-cycle cost	LC.	Including development, production and operation costs
Compactness	Comp.	Operation on board

Table 1: Design drivers

1.3 Rotor selection

The first task of choosing the rotor configuration was approached by listing the pros and cons of existing helicopters, from which the one that satisfies the design drivers for the proposed scenario was selected. The most important pros and cons considered are summarised in Table 2

Configuration	Pros	Cons
Single main rotor	High CP.	Low Comp.
Coaxial rotors	High M. and HP.	Low CP.
Synchropter	High M., HP. and Comp.	Low CP.
Lift/thrust compounded	High CP.	Low HP.
Tandem	Good HP.	High LC.

Table 2: Pros and Cons of rotor configurations

After group discussion, the synchropter rotor configuration is chosen. Single main rotor is excluded due to large power loss for tail rotor at low speed, which also lead to less compact design.

Tandem is excluded for the unnecessarily large fuselage. Compounded designs are also excluded as cruise performance is considered secondary from the mission definition. Although coaxial rotors and synchropters are very similar in terms of performance, the former is discarded due to the need for designing a sophisticated axle system with counter rotating concentric shafts, which also contribute to a large amount of drag in cruise. A modern synchropter, Kaman K-MAX is shown in Figure 1 for illustration

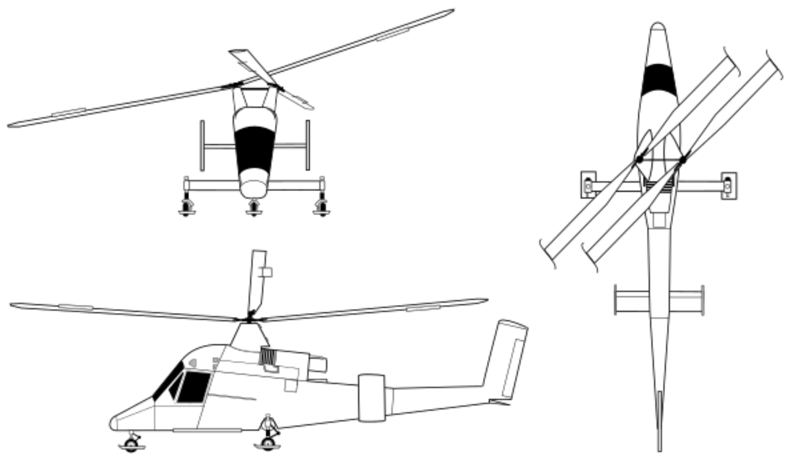

Figure 1: Kaman K-MAX

2 Empennage Configuration

Having chosen synchropter configuration, the empennage configuration becomes the next target. Similar to empennage on fixed wing aircrafts, the empennage of helicopters provides stability and control in pitch and yaw. However, the flow field around the helicopter empennage is more complicated than that of its counter part, the main reason being the effect of wake from the main rotor. This naturally post a question on the placement of empennage, most importantly the location of horizontal stabiliser.

2.1 Location

Three horizontal tail location was compared. Firstly, T-tail/high-mounted is discarded due to the need for a very stiff vertical stabiliser and low compactness. Between forward- and aft-mounted horizontal stabiliser, the former was chosen. A key difference between these two is the fact that forward-mounted location would lead to constant interaction between the rotor wake and horizontal surface even at low speed and hover. This is favourable in terms of control design, as there won't be a sudden change in download due to the boundary of the wake moving across the horizontal stabiliser when transiting from hover to forward flight (see Figure 2 for its counter part). Rotor-tail interaction is reviewed in preliminary design stage in Section 6.

Having chosen the horizontal tail configuration, an H-tail configuration is chosen for the vertical stabiliser to further enhance the effectiveness of vertical stabiliser with two vertical surface located at the ends acting as end plates. The flow field under the main rotor wake is complicated with spiral vortices and periodic vortex filament impinging on the horizontal stabiliser upper surface, creating pressure fluctuations and spanwise flows [4]. To have vertical surfaces at the tip of horizontal

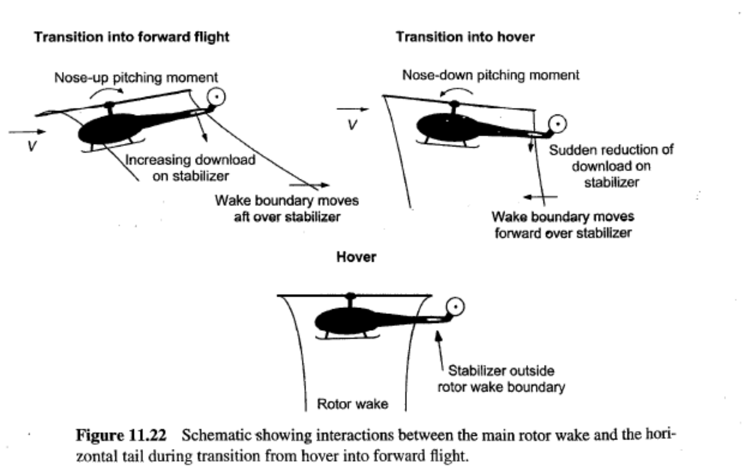

Figure 2: Rotor-tail interaction [1]

stabiliser acting as end plates, the complex flow field becomes more regular and the trailing vortices are damped.

The proposed empennage configuration can be visualised in Figure 3 below

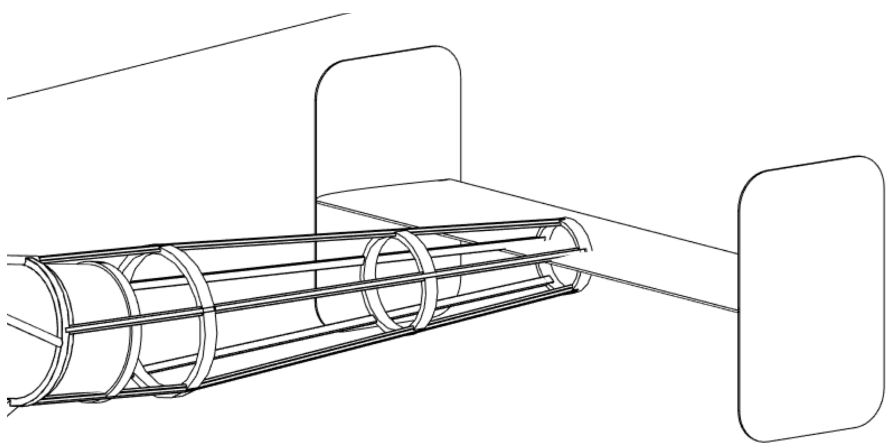

Figure 3: Empennage configuration (Courtesy of Mr. Malhotra, Tejasva [2])

2.2 Control surface

All-moving horizontal stabiliser, or stabilator, is chosen as method of control for the horizontal surface. Stabilator significantly improves the aerodynamic performance and handling quality due to mitigated rotor-tail interaction in various flight condition, which is achieved by the large range of tail pitching angle. However, this requires a careful controller design, with the tail plane setting angle as control input. There is also associated weight penalty due to extra pitching hinge mechanism, not to mention the two vertical stabiliser mounted at the ends. Nevertheless, the aerodynamic benefits is considered to exceed its structural counter part, as download on the horizontal surface due to rotor wake is very large in hover and low speed flights considering its relative location to the rotor wake region.

For the vertical stabiliser, conventional rudder control surface is chosen. This provides sufficient control in yaw with high structural efficiency.

Part II
Preliminary Design

Having finalised conceptual design, preliminary design was conduction after second week. The second part includes the aerodynamics analysis of rotor blade aerofoils, design of rotor tip, design and performance analysis of empennage, and aerodynamics analysis of the grapple device.

3 Rotor Aerofoils

Five aerofoils for the rotor blade design are analysed and compared using commercial software ANSYS Fluent. In the solution setup, Reynolds number is set to $Re = 10^6$ due to insignificant change in lift and drag coefficient at high Reynolds number. The variation in Mach number is taken into accounts by setting flow as air with Ideal gas assumption. The viscosity is changed accordingly to keep the Reynolds number constant at different Mach number. Sutherland's law is used to correct for Mach number effects on viscosity. Details of the mesh can be found in Appendix A.1. 2-D RANS with Spalart-Allmaras turbulence model is used to solve the problem.

3.1 Tip aerofoil selection

The aerofoil selection process for rotor blade tip is based on the aerofoils' L/D_{max} and M_{DD}, both directly affect the helicopter's hovering and cruise performance.

Considering the relatively high speeds at which tip section is operating, for the five aerofoils chosen, L/D_{max} at Mach number 0.7 was found by iteratively tracking down to the angle of attack that gives the maximum L/D. The results are shown in Figure 4.

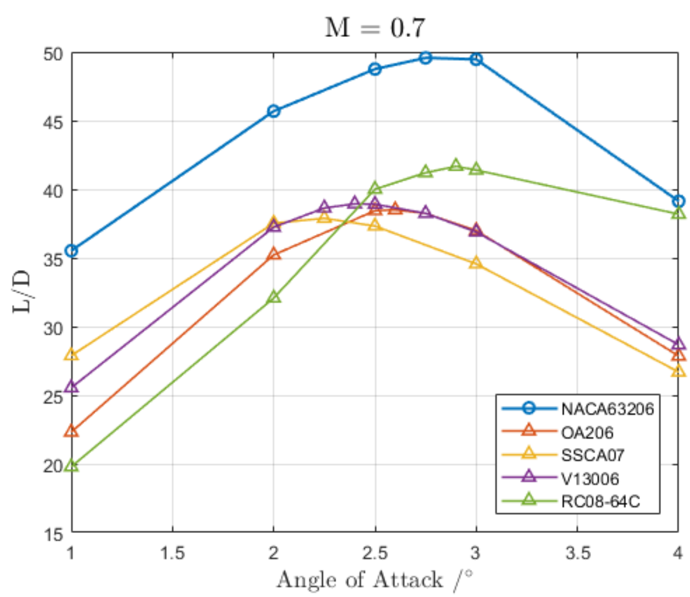

Figure 4: L/D_{max} for five aerofoil

Observing from the results, NACA63206 shows an exceptionally high L/D at the selected range of angles of attack. It is worth noting that, NACA63206 is a laminar aerofoil designed for wings of jet airliners, whereas the other four aerofoils are rotor-craft specific supercritical aerofoils. However, the operating condition of helicopter blades is drastically different from that of a wing. By having a synchropter design, the advancing side of the blade in cruise, which contribute to largest portion of lift, is above the retreating side of the blade from the opposite rotor due to rotor tilting in y-z plane. Thus, the advancing side should be clear from the wake of the other rotor and the inflow condition is less likely to be turbulent, of which the benefits of laminar aerofoil can be exploited.

Wave drag is only significant in high speed cruise when the blade tip could reach sonic condition. Thus, it is worth comparing the drag divergence Mach number of the aerofoils. For the ease of simulation, the angle of attack is set to be $\alpha = 2.5°$. Results of M_{DD} are summarised in Table 3 below

Aerofoil	M_{DD}
NACA63206	0.773
OA206	0.767
SSCA07	0.773
V13006	0.764
RC08-64C	0.770

Table 3: Drag divergence Mach number @ $\alpha = 2.5°$

Although NACA63206 is not designed specifically for reducing wave drag at high speeds, it has shown a relatively high M_{DD} when compared with other supercritical aerofoils. It was probably due to relatively small thickness to chord ratio that the difference in M_{DD} between laminar and supercritical aerofoil is very marginal.

Overall, NACA63206 is chosen for its high L/D_{max} and moderately high M_{DD} which maximise the hovering and cruise performance of rotor blade.

3.2 Aerofoil characteristics at various angles of attack

In order to define the blade geometry to an optimum [5], lift and drag polar at various angles of attack is required. The root section aerofoil is chosen to be NACA23012 by Alessandro [6]. Using the same CFD setup, the lift and drag polars for the two aerofoil at different Mach number are obtained through 520 cases of 2-D simulations. The results is shown in Figure 5 below.

A key limitation of RANS method is its inability to predict separation due to the fact that it is a transient process. For this reason, experimental values of lift and drag coefficients are used for angles of attack between 0° and 15° at these Mach numbers [7] [8].

4 Blade Tip Design

The blade tip design is achieved by studying past helicopters, in particular, the BERP-rotor shown in Figure 6. Two features are observed from these type of rotors - sweep and notch [3]. These geometric parameter are then applied to the Blade-Element-Momentum-Theory (BEMT) code by Chi Cheng [5] through modifications of local Mach number and chord length. However, 3-D effects such as vortex interactions introduced by the notch are not considered in the simplified BEMT model.

4.1 Sweep

Sweeping of the rotor tip section, similar to a swept wing, is employed to reduce wave drag at relatively high Mach number. From mission definition [5], the helicopter needs to be able reach a

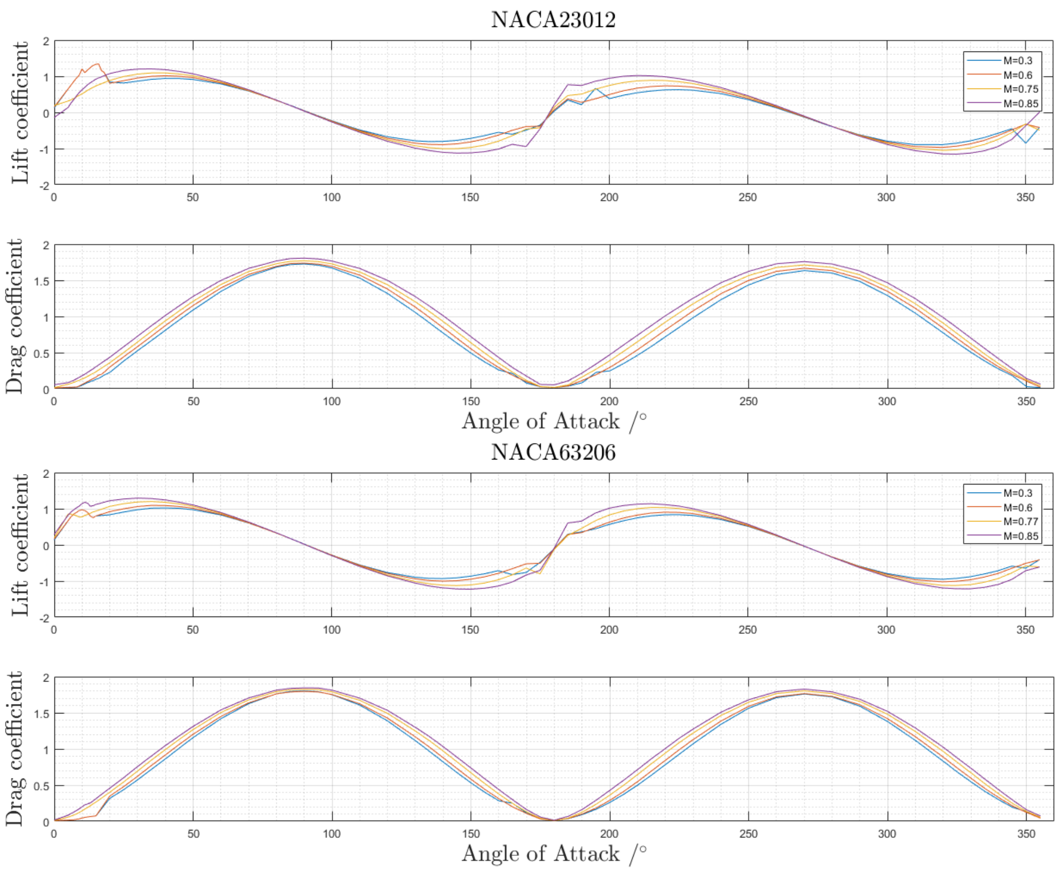

Figure 5: Lift and drag polar

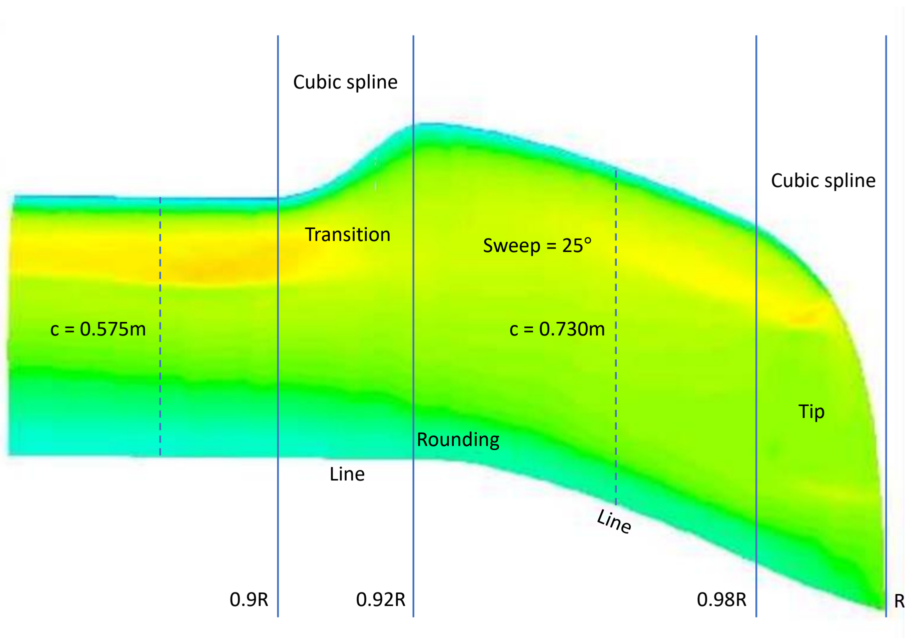

Figure 6: BERP Rotor tip (not to scale)

cruise speed of 225km/h when the rotor tip reaches Mach number 0.85. This is natural to define the sweep such that the relative Mach number reaches aerofoil's M_{DD} when the local Mach number reaches 0.85. The sweep angle is found to be $\Lambda = \cos^{-1}(M_{DD}/0.85) \approx 25°$. Note that the amount of sweep introduced is comparable to the example BERP tip employed on the Lynx [3].

4.2 Notch

The introduction of notch is to introduce a notch vortex at high angles of attack, which rotate in opposite direction to the tip vortex. The two vortices then generate a "downwash" on the suction surface of the tip section, which delays separation (See Figure 7). Although it would be favourable to obtain an optimal notch geometry using CFD, it was deemed impossible due to constraint on computational resource available and most importantly time limit. A reverse engineering approach is therefore employed, and the relative geometries are measured from a example BERP-rotor [3] shown in Figure 8.

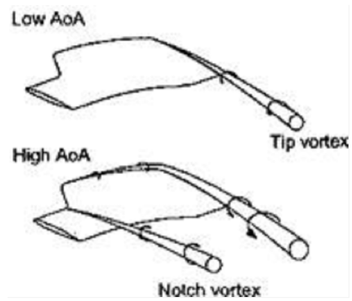

Figure 7: Notch vortices [3]

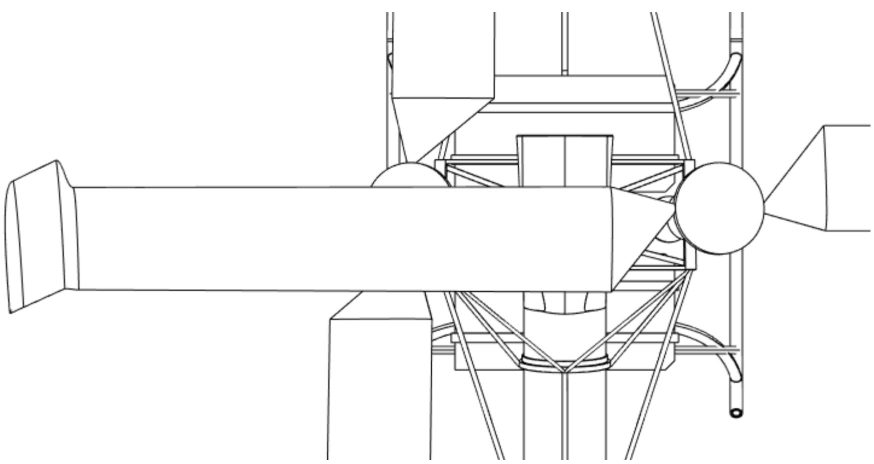

Figure 8: BERP Rotor tip design (to scale)

5 Empennage Sizing

Empennage design has been a well-known challenge in helicopter design due to complex rotor-tail interactions [1]. Luckily, by choosing synchropter configuration, tail rotor is not needed. However, the effect of main rotor is still present. This section present the process of sizing the horizontal and vertical stabiliser.

5.1 Horizontal stabiliser

Unlike horizontal stabiliser of fixed-wing aircraft that balance the pitching moments of wing and weight at CG, the design for helicopter has to consider the additional moment needed due to gyroscopic effect of the rotating main rotors. Therefore, instead of considering the tail volume coefficient, a new variable is introduced [9]:

$$K_H = \frac{\eta l_H S_H 10^3}{T \cdot h + K_r} \tag{1}$$

where η is an efficiency factor that accounts for the effects of the horizontal stabiliser relative location to rotor wake, with a typical value of 0.75 for a low horizontal stabiliser due to strong rotor-tail interaction. T is the main rotor thrust, h is the height of the main hub relative to the helicopter's centre of gravity, and l_H is the length of tail arm. The constant K_r takes into account the rotor moment contribution which may be written in the form:

$$K_r = \frac{l_\beta k M_s \Omega^2}{2} \tag{2}$$

where l_β is the offset of blade flapping hinge, k is the total number of blade, M_s is the blade moment of mass about flapping hinge and Ω is the rotor rotational speed.

Considering the application, K_H is chosen to be 3.5 for good stability and control in pitch [9].

5.2 Vertical stabiliser

The function of vertical stabiliser for helicopter is very much the same as that for a fixed wing aircraft - lateral and directional stability. As yawing moment generated by the vertical stabiliser is nearly parallel to the rotor axis, the effect of rotor in directional stability is negligible. Thus, it is natural to introduce a variable:

$$K_V = \frac{\eta l_V S_V 10^3}{T \cdot h} \tag{3}$$

where the effect of rotor K_r is omitted.

For utility helicopters, $K_V = 2$ provides sufficient lateral stability. Note that this value is for only one vertical surface of the two H-shape vertical surfaces.

In addition, a rudder chord ratio of 0.3 is chosen to ensure sufficient control in yaw.

5.3 Geometry

All inputs to equation 1, 2 and 3 are obtained from initial weight sizing and rotor sizing [10], which are summarised in Table 4 below.

Inputs	Value	Unit
l_H & l_V	13.2	ft
k	4	1
M_s	688.47	ft lbf
Ω	8.13	rev/s
T	6408.80	lbf
h	4.69	ft

Table 4: Input parameters for horizontal tail sizing

These input yield empennage platform areas of $S_H = 1.40m^2$ and $S_V = 0.57m^2$. To prodce more detailed geometry, variables such as aspect ratio and taper ratio needs to be defined. Unlike fixed wing aircraft, the shapes of helicopter empennages really form a cornucopia. For simplicity, a rectangular shape is chosen for both horizontal and vertical stabiliser to reduce the number of variables in preliminary design. The aspect ratio of the horizontal stabiliser is chosen to be $AR_H = 4$, which provides moderate stalling behavior at high angle of attack and low structural weight. The aspect ratio of the vertical stabiliser is chosen to be $AR_V = 1.5$ each, which leaves room for rudder control surface. Although these values seem to be determined arbitrarily, they are comparable to modern design such as Kaman K-MAX [11]. The span and chord length are calculated respectively and summarised in Table 5 below

Horizontal		Vertical	
s_H	2.37m	s_V	0.92m
c_H	0.59m	c_V	0.62m

Table 5: Empennage geometry

In terms of aerofoil choice, symmetric aerofoil NACA0012 is selected for both vertical and horizontal stabiliser due to the need to generate both positive and negative lift under different flight conditions [12].

6 Empennage Performance

Having obtained preliminary design of the empennage, the forces and moments contributions from the empennage are required by the Flight Mechanics and Control team for linearised model and controller design.

6.1 Forces and moments of horizontal stabiliser

The analysis of horizontal stabiliser start with finding the incidence angle. Following the convention of (u, v, w) being translational velocities and (p, q, r) being rotational velocities, the local incidence of horizontal tail can be expressed as

$$\alpha_H = \alpha_{H0} + \tan^{-1}\left(\frac{w + q(l_H + x_{CG} - w_i)}{u}\right), \tag{4}$$

where w_i is the induced velocity or downwash generated by the main rotor and α_{H0} is the control input setting angle of the horizontal stabiliser. A detailed analysis of downwash is presented in Section 6.3.

The magnitude of local velocity at horizontal tail is

$$V_H = \sqrt{u^2 + (w + q(l_H + x_{CD}) - w_i)^2} \tag{5}$$

Thus, lift generated by the horizontal tail is found as

$$Z_H = \frac{1}{2}\rho V_H^2 S_H C_{ZH}(\alpha_H) \tag{6}$$

where $C_{ZH}(\alpha_H)$ is the lift coefficient of the horizontal tail.

At small angle of attack, $C_{ZH}(\alpha_H)$ can be approximated with linear functions in the form

$$C_{ZH}(\alpha_H) = -a_{0H}\alpha_H \tag{7}$$

However, as the empennage experience a much wider range of angles of attackes, a more complex model is proposed [13]

$$|C_{ZH}| < C_{ZHl} \quad C_{ZH}(\alpha_{0H}) = -a_{0H}\sin(\alpha_H) \tag{8}$$
$$|C_{ZH}| > C_{ZHl} \quad C_{ZH}(\alpha_{0H}) = -a_{0H}\sin(\alpha_H)/|\sin(\alpha_H)| \tag{9}$$

where $C_{ZHl} = 2$ for NACA0012. The lift curve slope (a_{0H}) of horizontal stabiliser takes into account the end plate effects of the vertical stabiliser which was thoroughly investigated in a paper by Ashrafi [14]. For the current tail geometry, $a_{0H} = 4.8/rad$ is read from that paper.

The contribution to moment follows naturally from the force equation 6

$$M_H = (l_H + x_{CG})Z_H \tag{10}$$

The empennage drag contribution is considered to be negligible when compared with drag contribution from fuselage and rotor.

6.2 Forces and moments of vertical stabiliser

Similar to the horizontal tail, the vertical tail force analysis start by finding the sideslip angle or the local incidence, which can be expressed as

$$\beta_V = \beta_{V0} + \sin^{-1}\left(\frac{v - r(l_V + x_{CG})}{V_V}\right) \tag{11}$$

where the setting angle is assumed to be $\beta_{V0} = 0$. The magnitude of local velocity of vertical tail plane is

$$V_V = \sqrt{u^2 + (v - r(l_V + x_{CG}))^2} \tag{12}$$

Note that the "sidewash" generated by the two inter-meshing rotors is assumed to be cancelled.

Then the side force is

$$Y_V = \frac{1}{2}\rho V_V^2 S_V C_{YV}(\beta_V, \delta_r) \tag{13}$$

where the lift coefficient C_{YV} is a function of the sideslipe angle β_V and rudder deflection angle δ_r. Note that S_V is the total area of two vertical stabilisers in this equation.

Similar to the vertical tail, at small sideslip angle C_{YV} can be approximated by linear functions of the form

$$C_{YV} = -a_{0V}\beta_V + a_r\delta_r \tag{14}$$

Unlike horizontal stabiliser, the precise performance of vertical stabiliser is more complicated due to its low aspect ratio. A more complex model used for analysis of SA 330 helicopter with higher order polynomial is proposed [15] [13],

$$C_{YV} = -a_{0V}(11\beta_V^3 - 85\beta_V^5) + a_r\delta_r \tag{15}$$

where $a_{0V} \approx 3/rad$ is estimated due to relatively low aspect ratio and $a_r \approx 0.008/°$ [14].

The moment contribution from the vertical stabiliser can be found easily

$$N_V = -(l_V + x_{CG})Y_V \tag{16}$$

Note that the moment contribution from increased camber due to rudder deflection is comparatively small, thus ignored.

Due to different complexity, Flight Mechanics and Control team has the freedom to choose which model of lift coefficient being used depending on the flight condition.

6.3 Downwash model

In Section 6.1, the angle of attack of the horizontal stabiliser is expressed as a function of the downwash velocity. A preliminary model of downwash velocity is constructed and presented here.

In forward flight, the downwash velocity at the rotor can be found by

$$w_r = \Omega R \frac{C_T}{\sigma} \frac{\sigma}{2\mu} \tag{17}$$

where Ω is the rotor rotational velocity, R is the rotor radius, C_T is the thrust coefficient, σ is the rotor solidity, and $\mu = u/\Omega R$ is the advance ratio. Note that, C_T is dependent on altitude due to change speed of sound that leads to change in local Mach number, therefore different efficiency. Thanks to my colleague, the data for downwash velocity in forward flight is readily obtained for various advance ratio and altitude [16].

Due to the assumptions made for forward flight, the downwash velocity in equation 17 blows up when the forward velocity is zero. Therefore, the data of downwash velocity in hover is calculated separately [16] and interpolated with data in forward flight to generate 2-D linear interpolation of downwash velocity as a function of advance ratio and altitude, i.e. $w_r = w_r(\mu, alt)$.

When determining the location of empennage in Section 6.1, the shape of the wake region (in hover) was assumed to be a straight cylinder. However, it becomes clear that the wake region contracts downstream of the rotor, which leads to an amplification of downwash velocity. The initial location of the tail therefore place the horizontal tail outside the wake region in hover and low-speed flights. According to my colleague's report on fuselage [6], the downwash velocity at the fuselage and horizontal tail is approximately 1.3 times that at the rotor plane, i.e. $w_i = 1.3w_r$.

As the location of horizontal tail is right under the rotor tip, the rotor downwash would not interact with the rotor in hover and low-speed forward flight. A model of the downwash shown in Figure 9 is proposed to incorporate the effect of wake boundary moving across horizontal tail during transition from low-speed to high-speed forward flight. The downwash is assumed to be constant across blade span.

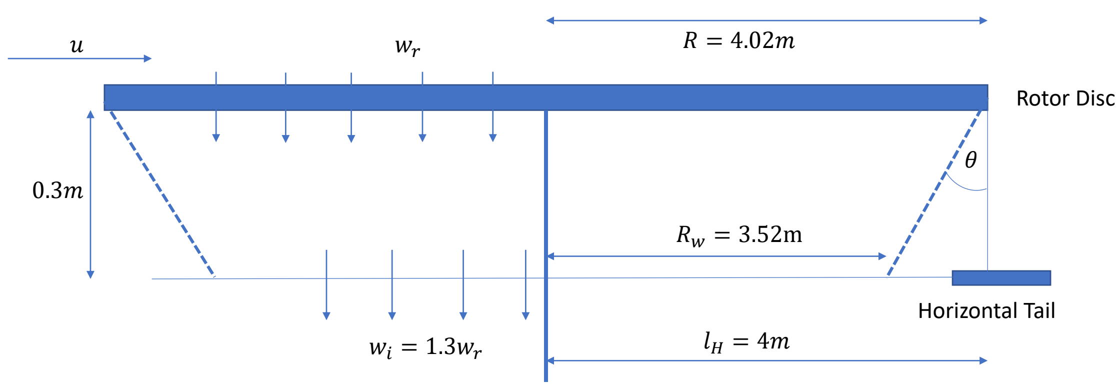

Figure 9: Downwash model (not to scale)

Firstly, a simplified control volume consisting of the wake boundary of a rotor, the rotor disk, and the "wake disc" at the plane of horizontal tail is considered. The effect of having two rotors are considered as superposition of downwash velocity. By applying continuity and assuming incompressibility, the radius of the wake disc R_w is calculated as

$$w_r \pi R^2 = w_i \pi R_w^2 \longrightarrow R_w = 3.52m \tag{18}$$

Then from trigonometry, a wake contraction angle $\theta \approx 60°$ is calculated. Now, considering the effect of forward velocity u as to advect the entire control volume downstream, a wake advection angle can be calculated as

$$\theta_w = \tan^{-1}\left(\frac{u}{1.15w_r}\right) \tag{19}$$

where the 1.15 factor is the average between w_r and w_i.

Finally, a condition is set in MATLAB such that if $\theta_w < \theta$, then $w_i = 0$. The MATLAB function "downwash.m" can be found in Appendix B

7 Grapple Device Drag

Towards the end of the project, a drag analysis of the grapple hook is required by the System team [17]. ANSYS Fluent is used again to find the drag of the grapple device with respect to zero flap deflection angle. Through an iterative process, the final grapple device geometry is a near square vertical fin with 1m chord length and 1.08m span. 3-D RANS with Spalart-Allmaras turbulence model is employed to find a drag coefficient of 0.01239 with zero flap deflection angle. Details of the mesh can be found in Appendix A.2

8 Discussion

Having completed conceptual and preliminary design of the helicopter, a final review is made to direct furure developments together with potential improvements.

Regarding the aerofoil aerodynamics characteristics analysis, the prediction of stall has been at the forth front of academic research. In 2-D analysis, transient simulation would provides insight into the onset of separation. Structured mesh should also be considered for better convergence of the results.

So far, only 2-D aerofoil analysis was conducted for the blade. The performance of the rotor blade in hover and cruise is obtained using BEMT developed by my colleagues which ignores 3-D effects [5] [16]. Future steps toward refinement and verification could be a fully 3-D simulation of the rotor blades and potentially further optimisation with respect to the effect of spanwise flow. In particular, a 3-D CFD analysis of the notch feature would provide information for the separation control of the tip section at high angle of attack.

The sizing of the empennage was based on historical data. However, a refinement can be made together with Flight Mechanics team by plotting the root locus diagram of stability matrix for different area or tail arm length of the empennage, from which an optimal value may be extracted.

Comming back to the aerodynamics analysis of downwash, the present model is rather coarse. Firstly, the boundary of the wake should not be straight. The shape of control volume should be similar to that in Figure 2. Secondly, the assumption that the downwash velocity being constant along blade span is not valid. There definitely exists a variation of downwash velocity along the span and also due to advection. Thirdly, the effect of inter-meshing rotors is only considered as a superposition of the downwash velocity. Further CFD analysis on the shape and structrue of the flowfield within the rotor wake region should be considered.

Finally, the aerodynamics coefficients of the empennage, as functions of incidence angle and rudder deflection, could have been obtained from CFD analysis. However, without a proper rotor wake analysis, an accurate prediction of lift and drag of empennage is impossible.

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A Appendices

A.1 Mesh Details for 2-D Aerofoil Aerodynamics Analysis

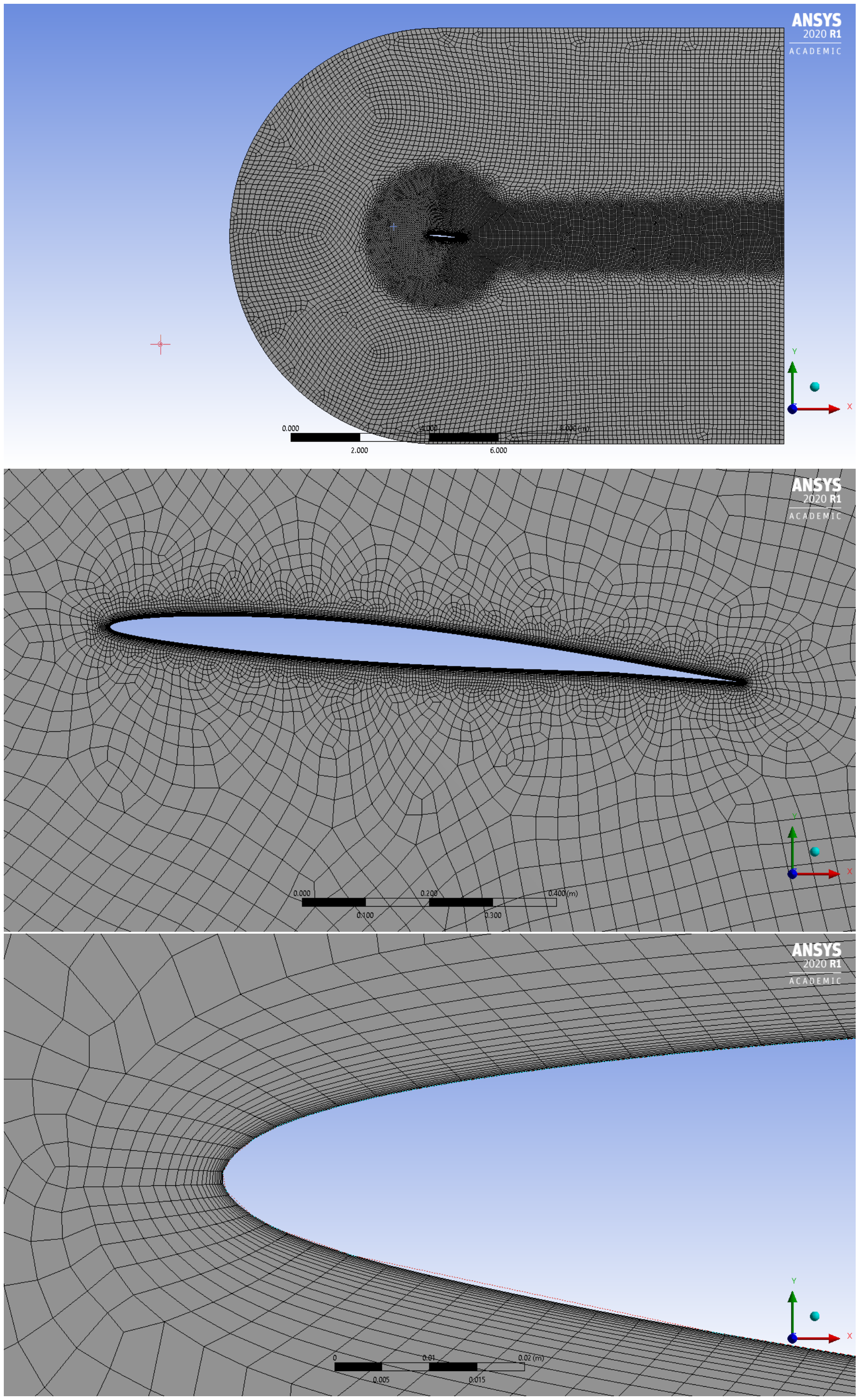

Figure 10: Mesh of NACA63206 at $\alpha = 10°$

A.2 Mesh Details for 3-D Grapple Hook Aerodynamics Analysis

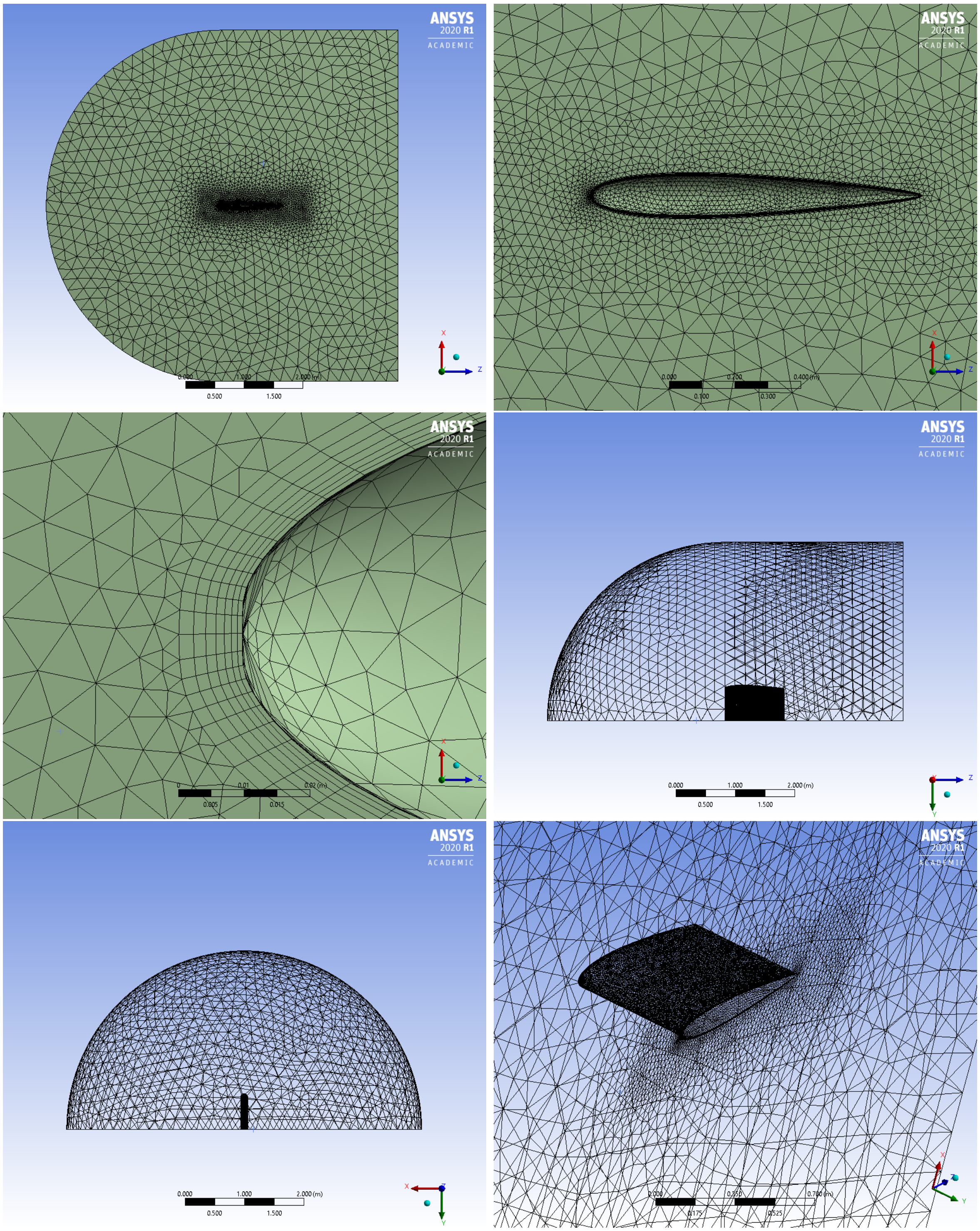

Figure 11: Mesh of Grapple Hook

B MATLAB code for downwash

This script calculate downwash velocity for the horizontal tail at different advance ratio and altitudes.

```
% AERO96005 GROUP DESIGN PROJECT
% Le Yin — Aerodynamics sub team

%This code calculate the downwash w_i experienced by the horizontal tail
% at different altitude and advance ratio.
function [w_i] = downwash(mu,altitude) %altitude in ft
%% Loading Data
dw_h = ones(5,1)*15.54; %Hovering downwash at rotor tip section
load('ctsigma_vs_mu.mat'); %Load data provided by Enson for forward flight
sigma = 0.0909;
omega = 8.125*2*pi;%rad/s
R = 4.02; % radius of rotor in meter
[ar,¬] = meshgrid(advanceratio,alt);
dw = omega*R*sigma/2*meanCTsigma./ar;

%% Introducing hover downwash
advanceratio = [0 advanceratio];
dw = [dw_h dw];
[advanceratio,alt] = meshgrid(advanceratio,alt);
w_i = 1.3*interp2(advanceratio,alt,dw,mu,altitude); %induced velocity
%at horizontal tail plane

%% Check if tail is under rotor wake
%Wake contraction at tail
w_ir = w_i/1.3; %induced velocity at rotor
Rt = R/sqrt(1.3); %Radius of wake region using continuity
lt = 4;  %distance of tail from rotor hub
zt = 0.3; %0.3m below from rotor
u = mu*omega*R; %forward velocity
a1 = atan((R—lt)/(zt));
a2 = atan((R—Rt)/(zt));
amin = a2—a1; %angle of wake
if atan(u/1.15/w_ir)<amin
    w_i = 0;
end
```